Alex Chen

Object Detection · Computer Vision Engineer

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Open to relocation: Germany · Netherlands · EU | Eligible for EU Blue Card

Professional Summary

Computer vision engineer with an M.Eng. in Computer Science and 2+ years of research focused on real-time object detection and multi-scale feature fusion. Designed and deployed two detection architectures — an attention-enhanced YOLO variant and a transformer-CNN hybrid detector — achieving state-of-the-art mAP on COCO and domain-specific industrial datasets. Experienced in edge deployment (TensorRT / ONNX) and building end-to-end perception pipelines from data annotation to production inference.

Education

M.Eng. in Computer Science *Sep 2022 – Jun 2025*

University of Science and Technology, China

Thesis: *Multi-Scale Attention Networks for Real-Time Object Detection* | Advisor: Prof. Wei Zhang

- Proposed two novel detection architectures; benchmarked on COCO, VOC, and custom industrial datasets.
- Trained on NVIDIA A100 cluster with PyTorch; implemented full ablation studies and latency profiling.

B.Eng. in Software Engineering *Sep 2018 – Jun 2022*

National University, China

Thesis: *Pedestrian Detection in Crowded Scenes via Occlusion-Aware Feature Learning*

Technical Skills

Detection & Vision:	YOLO (v5/v7/v8), Faster R-CNN, DETR, RT-DETR, Anchor-Free Detectors, Feature Pyramid Networks, NMS variants
Deep Learning:	PyTorch, CNN, Vision Transformer, Attention Mechanisms, Knowledge Distillation, Quantization-Aware Training
Deployment:	ONNX, TensorRT, OpenVINO, Triton Inference Server, CUDA, C++ inference pipelines
Engineering:	Python, C++, OpenCV, NumPy, Docker, Git, Linux, MLflow, Weights & Biases
Data:	COCO, VOC, LVIS, Roboflow, CVAT annotation, synthetic data augmentation (Albumentations, Mosaic)
Languages:	Mandarin (native), English (fluent), German (A2)

Research & Engineering Projects

EfficientDet-SA — Scale-Aware Attention for Real-Time Object Detection *2024*

- Designed a scale-aware attention module that dynamically re-weights multi-scale features in the FPN neck, improving small-object detection by 3.2 mAP on COCO `val2017`.
- Introduced a lightweight channel-spatial gating unit (0.8M extra params) compatible with YOLO and EfficientDet backbones.
- Achieved 42.6 mAP at 58 FPS on RTX 3090 — Pareto-optimal on the accuracy-latency frontier vs. YOLOv8-M and RT-DETR-R50.

Tech: PyTorch, CUDA, Albumentations, Weights & Biases, A100 GPU

HybridDet — Transformer-CNN Hybrid Detector for Industrial Defect Inspection *2024 – 2025*

- Proposed a hybrid architecture pairing a Swin-Tiny backbone with a CNN detection head for high-resolution industrial images (4K).
- Deployed via TensorRT FP16 on Jetson Orin; reduced inference latency from 48ms to 11ms while maintaining 91.3% mAP on a proprietary PCB defect dataset (12 classes, 50K images).
- Open-sourced training pipeline and ONNX export scripts; 200+ GitHub stars.

Tech: PyTorch, TensorRT, ONNX, Jetson Orin, Docker

Autonomous Driving Perception — 3D Object Detection Module *2023 – 2024*

- Contributed to a LiDAR-camera fusion pipeline for 3D vehicle detection; implemented BEV feature alignment module.
- Improved nuScenes NDS by 1.8 points over BEVFusion baseline through temporal feature aggregation.
- Built automated evaluation CI/CD with MLflow tracking and nightly regression on 8×A100 cluster.

Tech: MMDetection3D, BEV Fusion, nuScenes, MLflow, Kubernetes

Real-Time PPE Compliance Detection System — Construction Sites *2023*

- Built an end-to-end safety helmet and vest detection system processing 16 RTSP camera streams simultaneously.
- Fine-tuned YOLOv8-S on 8K annotated images; achieved 94.7% mAP@0.5 with <15ms per frame on RTX 3060.
- Deployed as a Docker microservice with REST API; integrated with client's alert dashboard.

Tech: YOLOv8, OpenCV, FastAPI, Docker, Redis

Funded Research Participation

Small Object Detection in UAV Imagery Under Complex Backgrounds *Provincial S&T Project*

Grant No.: XXXX-2024-XXXXX | Role: Core Participant

Multi-scale feature fusion and context-aware attention for drone-captured aerial object detection.

Vision Foundation Models for Industrial Quality Inspection *National Youth Fund*

Grant No.: 62XXXXXXXX | Role: Participant

Pre-training and fine-tuning large vision models for zero-shot and few-shot defect detection.

Professional Experience

Computer Vision Engineer — Part-Time Contract *2024 – Present*

Developed and maintained object detection pipelines for a logistics automation startup. Responsibilities include model training, A/B testing detection architectures, and deploying TensorRT-optimized models on edge devices. Reduced package misclassification rate by 37% through a custom cascade detection strategy.

Awards & Recognition

- **Gold Award** — Provincial "Internet+" Innovation & Entrepreneurship Competition, 2024.
- **Silver Award** (×2) — Provincial "Internet+" Innovation & Entrepreneurship Competition, 2023.
- **Second Prize** — National Artificial Intelligence Challenge (Object Detection Track), 2023.
- **First-Class Academic Scholarship** — University of Science and Technology, 2022–2025.
- **National Encouragement Scholarship** — Ministry of Education, China.

Beyond Work

Passionate about the intersection of perception and robotics — contributing to open-source detection toolboxes and writing technical blog posts on real-time inference optimization. Active participant in Kaggle competitions (Expert tier). Planning relocation to Europe for industry R&D roles in autonomous systems or industrial AI.